

Hope Artificial Intelligence

Assignment - Regression Algorithm

Problem Statement or Requirement:

A client's requirement is, he wants to predict the insurance charges based on the several parameters. The Client has provided the dataset of the same.

As a data scientist, you must develop a model which will predict the insurance charges.

1.) Identify your problem statement

client wants to predict **medical insurance charges** for individuals based on several personal attributes:

age, sex, BMI, number of children, and smoking status. Since the target variable (*charges*) is a continuous

numeric value, this is a **supervised regression problem**. The objective is to build multiple regression

models, evaluate each using the **R2 score**, document the results, and select a final model best suited to

predict insurance charges accurately.

2.) Tell basic info about the dataset (Total number of rows, columns)

I Total rows: **1338**

I Total columns: **6** — age, sex, bmi, children, smoker, charges

I Target column: **charges** (continuous)

I No missing / null values in any column

I Two nominal (string/categorical) columns: **sex** (male/female) and **smoker** (yes/no)

3.) Mention the pre-processing method if you're doing any (like converting string to number - nominal data)

The dataset contained two **nominal (string) columns** that needed to be converted to numeric form

before modeling:

I **sex**: male ® 0, female ® 1

I **smoker**: no ® 0, yes ® 1

As both columns had only two possible values, a simple binary label mapping was used instead of

one-hot encoding (which would add unnecessary columns for a binary feature). No other pre-processing

was required: there were no missing values and no outlier-removal was applied since the high-charge

cases are genuine, meaningful data points (driven mainly by smoking status), not data errors. Features

were split 80% train / 20% test, and standard scaling was applied only for the distance-based models

(SVR, KNN) which are sensitive to feature scale; tree-based models used the raw, unscaled features.

4.) Develop a good model with `r2_score`. You can use any machine learning algorithm; you can create many models. Finally, you have to come up with final model.

Four algorithms using below

1. multi linear regression
2. support vector machine
3. decision tree
4. random forest

5.) All the research values (r2_score of the models) should be documented. (You can make tabulation or screenshot of the results.)

Sl no	Model	R_square value	Result=array
1	Multi linear regression	0.7894	2560430.62605342
2	Support vector machine	0.810	14948.961940
3	Decision tree	0.739	36580.28216
4	Random forest	0.849	27994.4061844

6.) Mention your final model, justify why you have chosen the same.

My final model is random forest.

It achieved the highest R2 score (0.849).

Comparing other models this is highest r2 score